Making the Intraoperative Count Exact: Anchored Overlapping Fluoroscopy for Vertebral Level Confirmation Without a Learned Component

Gregory Schwing, MD, PhD[1]

[1]Department of Surgery, Detroit Medical Center / Wayne State University School of Medicine, Detroit, MI

Corresponding author: G. Schwing, [redacted]

Article type: Original Research

Manuscript word count (Introduction–Discussion): ≈2,900

Tables: 3 **Figures:** 0 **Funding:** none declared.

Conflicts of interest: the author has a commercial interest in a level-confirmation product under development.

Data sharing: all imaging is public and de-identified. Every counting arm reported here is written by one tool into a result file, and the table of arms is generated from those files rather than typed.

Abstract

Background: The Universal Protocol verifies a mark. A vertebral level carries no mark: it is counted intraoperatively across several limited fluoroscopic fields, and each join between fields can be one body off. Chart review put the protocol's ceiling at 62 % of wrong-site cases prevented, and a 34,544-case registry put wrong-level incidence at 0.12 %, rising to 0.42 % in the thoracic spine.

Purpose: To measure whether overlapping exposures linked from an anatomical anchor make the intraoperative count exact, and to locate the residual risk that remains when it is.

Materials and Methods: Retrospective study of public labelled CT and radiograph collections. Frames spanning a 9-inch image-intensifier field (14 cm at the spine) were cut from each labelled long image by a fixed protocol—anchor at the edge of the first field, then a move of one field minus two vertebral bodies—and degraded independently. Consecutive frames were linked by normalised cross-correlation restricted to within 0.6 of a vertebral body of the step the protocol commanded. The target is the k-th node from the anchor. No learned component takes part in the count and no level name is formed, so no spine was withheld. Walks were run with the annotation's vertebra boxes and, separately, with a detector's boxes. The endpoint is the share of walks putting the mark on the body truly k from the anchor, among walks whose target entered a field before the image ran out.

Results: With the annotation's boxes, counting down from C2 at three bodies of overlap was exact on 834 of 834 walks across 76 whole-spine studies at a mean of 2.24 exposures, and on 2,646 of 2,654 reached walks (99.70 %) across 204 studies from an institution none of this work trained on. Counting up from the sacrum was exact on 6,234 of 6,255 reached walks (99.66 %) across all 801 abdominopelvic spines and on 7,168 of 7,228 (99.17 %) across 522 studies. On 16 real lateral radiographs reaching the sacrum, every walk was exact (95 of 95) at 1.61 exposures. The residual risk is in the boxes. On 79 spines held out from a render-trained detector, the annotation's boxes gave 618 of 618 reached walks exact; the detector's own boxes gave 187 of 622 (30.06 %), and the node the count started from was the true sacrum on 42.44 % of walks.

Conclusion: Counting across overlapping fields is exact and needs no network. What needs a network is naming, and naming is where the uncertainty lives.

Summary Statement: Overlapping fluoroscopic exposures linked from an anchor made the intra-operative vertebral count exact on 99.17–100 % of reached walks in nine arms of 16 to 801 studies each, using the annotation's vertebra boxes and with no learned component in the count; with a detector's boxes on the same spines it fell to 30.06 %, placing the whole residual risk in anchor detection.

Introduction

The Universal Protocol works by marking. The site is marked while the patient is awake, the mark is confirmed at the time-out, and the operation proceeds where the mark is. A vertebral level has no mark. It lies under several centimetres of tissue, it is indistinguishable from its neighbours, and it is identified after the incision, by counting on fluoroscopy from whatever landmark is visible [1, 2]. That is not a lapse in discipline; it is a step the protocol does not cover. Chart review of insured wrong-site cases judged that the Universal Protocol would have prevented eight of 13 under ideal conditions [3], wrong-site events continued at a steady rate through the protocol era [4], and the only systematic review to assess site-verification protocols in the spine found they failed to prevent about one-third of cases [5]. The remedy the spine literature recommends—imaging after exposure against a marked fixed structure [6]—is itself another count.

The failure is rare per procedure and common per career. A prospectively collected registry of 34,544 degenerative spine operations recorded wrong-level surgery in 0.12 % (95 % CI 0.09–0.16), with the thoracic spine highest at 0.42 % (CI 0.16–1.08), three times the lumbar and eight times the cervical rate [7]; anonymous self-report gives one in 3,110 procedures [8]; and about half of responding neurosurgical spine surgeons in two independent surveys report at least one wrong-level operation in their career [8, 9]. Seventeen percent of self-reported events led to legal action or settlement [8]; of 68 litigated wrong-disc-level cases, 55 ended in a payment and surgeons prevailed in 42 % of those a jury decided [10]; the median indemnity for a surgical never event reported to the National Practitioner Data Bank is $75,220 (IQR $28,847–$215,003, 2025 dollars) [11]; 978 English spinal-surgery negligence claims over five years were valued at £535.5 million [12].

Aggregate burden is bounded rather than known, and this paper declines to overstate it. The Joint Commission received 127 wrong-surgery reports in 2024, nine of them wrong-level and seven of those spinal, from a voluntary system it states supports no frequency conclusion [13]; NHS England recorded 185 wrong-site never events in one year [14]. Claims-based modelling puts surgical never events near 4,000 a year in the United States [15], and applying the two per-procedure rates above to a United States volume of 1.3–1.6 million spine operations gives roughly 400 to 1,900 wrong-level operations and, with the verified litigation and payment fractions

and a mean paid claim near $133,000–$136,000 [15, 16], on the order of $8–37 million a year in indemnity alone. Each step of that is an extrapolation, the spine-surgery denominator is unverified, and hospital and disability costs have no published per-event figure. The often-quoted $1.3 billion is a 21-year cumulative for all surgical never events [15], not an annual wrong-site figure, and no defensible worldwide figure exists.

What is spent avoiding the error is easier to price. One quaternary centre placed 179 preoperative percutaneous gold fiducials over 38 months, 176 of them thoracic, at a mean of 51.9 minutes and 183 mGy each, purely to mark a level for the surgeon [17]. At a hospital operating-room cost of $34–37 per minute before physician fees [18], that is roughly $1,750–$1,950 of staffed procedural time per patient and about 155 hours across the series—a proxy, since the marker is placed in a fluoroscopy or CT suite rather than a theatre. The fiducial buys real time: thoracic localisation falls from about 15 minutes to about three [19, 20], at the cost of a separate 14.3-minute CT intervention [21].

Automated alternatives have been built almost entirely on registering a preoperative CT to the intraoperative radiograph [22, 23]. Registration does not name a vertebra. It solves a six-degree-of-freedom pose and transfers a name a human read off the preoperative volume, which means it is unavailable when there is no CT, and a miscounted CT propagates onto the radiograph faithfully.

This paper starts from what surgeons actually do. They do not classify a vertebra by its shape; they count, from C2 down or the sacrum up, and the count is unreliable only because it is carried across several limited fields whose joins are ambiguous in a nearly periodic column. The intervention that follows is not a classifier that asserts an identity but a mechanism that makes the join exact. We measure how exact, where it breaks, and what it costs to run.

Materials and Methods

This retrospective study used public, de-identified imaging only; institutional review determined it not to be human-subjects research.

Imaging

Four sources were used. CTSpinoPelvic1K supplied 801 abdominopelvic CT studies with a per-level vertebral label [24]; VerSe supplied 76 whole-spine studies [25]; TotalSegmentator's public release of routine clinical CT supplied 204 studies holding C2 and 522 reaching the sacrum, from an institution on which nothing in this work has been trained; and real lateral radiographs supplied 16 spines reaching the sacrum and 24 holding C2. Serial sweeps of one patient do not exist in any public collection, so each sweep was cut from a single labelled long image. Every frame was degraded independently—gain, offset, gamma, a small rotation and scale, and a fitted film-noise model—so no two frames share pixels exactly.

The sweep protocol

The protocol is the one a radiographer already performs by hand, made explicit [26]. The anchor is placed at the edge of the first field: the sacrum at the bottom for lumbar and thoracolumbar work, C2 at the top for cervical and upper thoracic work. The C-arm then moves by one field minus two vertebral bodies, and the exposure is repeated to the level of interest. The last move is a full move or none. Frame span is physical rather than a count of vertebrae: a 9-inch image intensifier spans about 14 cm at the spine once the fixed 100 cm source-to-detector distance of a mobile C-arm is accounted for [27], which holds about four lumbar bodies.

Linking, and the count

Consecutive frames are linked by normalised cross-correlation on the overlap band alone, never by a global homography, and the search is restricted to within 0.6 of a vertebral body of the step the protocol commanded. That restriction is the mechanism of the paper. An unrestricted correlation on a nearly periodic column locks one body off, which is the documented failure mode of automatic vertebral localisation and accounted for all 47 failures in the original three-dimensional to two-dimensional registration study [28]. Restricting the search to the commanded displacement removes the aliasing, because the protocol states what the answer should be to within half a body.

The linked frames form one chain of vertebral nodes. A walk starts at the anchor node and the target is the k-th node from it. Nothing is fitted anywhere in this procedure: no encoder is

loaded, no level name is formed, no alphabet is chosen, and variant anatomy therefore never enters. Because nothing is fitted, no spine is withheld on a training-distribution argument, and every study carrying the anchor was used.

Box sources

The chain's nodes are vertebra boxes. Headline arms use the annotation's boxes, which isolates the linker: the only thing that can fail is the join. A second set of arms substitutes a detector's boxes and is restricted to spines held out from that detector. In the arms reported here the anchor node is the terminal node of the chain at the anchor end; identifying it by class instead is a separate arm that has not returned.

Endpoints

Reached is the number of walks whose target entered a field before the image ran out; a walk that does not reach is a limitation of field, not a counting error, and both denominators are reported. *Exact of reached* is the share of reached walks placing the mark on the body whose position in the labels is truly k from the anchor. *Exposures* is the mean number of frames a walk consumed. *Anchor held* is a diagnostic—how often the node the count began from really was the anchor—and is never used as a gate. *Refused* counts walks the system declined. Wrong walks are reported with the number of distinct spines they fall in, because a single bad study produces many bad walks.

Naming, and timing

For the division-of-labour comparison, the same chains were named by a ResNet-18 encoder over fixed-scale vertebra crops with a bidirectional recurrent head over the chain [29, 30]. Per-frame encoder time, per-walk linking time and per-walk decode time were measured in the same runs. No hypothesis test was applied; proportions are reported with their numerators and denominators.

Results

The count is exact, and it needs no network

Table 1 gives every counting arm. With the annotation's boxes, counting down from C2 at three bodies of overlap was exact on 834 of 834 walks over 76 whole-spine studies, at a mean of 2.24 exposures; at two bodies, 832 of 834, the two failures in one study. On 204 TotalSegmentator studies holding C2 the same protocol was exact on 2,646 of 2,654 reached walks (99.70 %) at two bodies and 2,640 of 2,650 (99.62 %) at three, the misses confined to two spines in each arm. Counting up from the sacrum was exact on 6,234 of 6,255 reached walks (99.66 %) over all 801 abdominopelvic spines at two bodies of overlap and 6,214 of 6,229 (99.76 %) at three, with errors in six and five spines; on 522 TotalSegmentator spines, 7,168 of 7,228 (99.17 %), the errors in six spines. Every arm consumed between 1.15 and 2.96 exposures.

The errors are not scattered. They sit far from the anchor. Counting up from the sacrum on the abdominopelvic spines was exact on 99.73 % of lumbar targets and 95.45 % of thoracic ones at two bodies of overlap, and on the TotalSegmentator sacrum arm the cervical targets—18 or 19 bodies from the anchor—fell to 83.84 % against 99.33 % lumbar. Counting down from C2 on the whole-spine studies shows the same gradient from the other end: 99.46 % thoracic at two bodies of overlap and 100 % at three. A count degrades with the number of joins it crosses, exactly as a surgeon's does; the remedy is to anchor at the near end, which the protocol permits because both ends of the column carry an unmistakable landmark.

What did not reach is reported separately. Ninety-seven of 6,352 walks on the abdominopelvic spines and 168 of 7,396 on TotalSegmentator ended before the target appeared, almost all of them the topmost labelled body or a cervical level far from the sacrum. Those are field limitations and are not counted as errors.

On real radiographs

On 16 real lateral radiographs reaching the sacrum, every walk was exact with the annotation's boxes: 95 of 95 reached, at 1.61 exposures. On 24 real radiographs holding C2, 110 of 110, at 1.15 exposures. Sixteen and 24 spines are thin, and the number is stated beside the percentage

for that reason. These two arms establish that the linker survives real radiographic pixels; they do not establish a rate.

The residual risk is the anchor, not the count

Table 2 is the result that governs deployment. On 79 CTSpinoPelvic1K spines held out from a render-trained detector, the annotation's boxes gave 618 of 618 reached walks exact. The detector's own boxes on the same spines gave 187 of 622 (30.06 %), with errors in all 79 spines; at three bodies of overlap, 178 of 621 (28.66 %) in 75 spines. The mechanism is visible in the diagnostic: the node the count started from was the true sacrum on 42.44 % of walks at two bodies and 39.77 % at three. A missed or spurious box shifts the ordinal, and shifting the ordinal renames every body above it. That is the surgeon's own error, reintroduced by a machine, and it is why the count alone is not a product.

The same 16 real films run with the deployed film detector's boxes gave 95 of 95 exact walks at 1.47 exposures while the anchor diagnostic read 93.68 %: some walks were scored as beginning from a node that was not the true sacrum and nonetheless ended on the correct body. Those two numbers are not reconciled here. Until they are, that arm is not evidence that a detector's boxes are safe; the 79-spine CT arm is the arm to believe.

Three arms address this directly and none has returned. Identifying the anchor by class rather than by chain position, on the CTSpinoPelvic1K detector boxes at two and at three bodies of overlap, and on the real films, is **[PENDING: anchor-by-class counting arms]**. A detector holding an explicit C2 class is **[PENDING: C2-class anchor detector]**. Confirming a level by re-imaging with a radiopaque instrument in the field and placing that instrument on the chain is **[PENDING: needle confirmation]**; the evidence brief carries the heading and no result.

One further gap is reported rather than explained: the refusal count is zero in every arm in Table 1 and Table 2. The system's declared refusal behaviour—declining an uncertain link, or a later exposure it cannot place—never fired in these runs. Refusal is therefore specified and unmeasured, and a system whose safety argument rests on refusing when it cannot see the anchor has not yet demonstrated that it refuses.

Naming is the part that is uncertain

The same chains named by the encoder and its chain decoder were correct on 96.25 % of 950 walks on CTSpinoPelvic1K and 94.69 % of 7,405 walks on TotalSegmentator, the latter corpus never trained on anywhere in this work. By region on TotalSegmentator: 96.21 % lumbar, 91.58 % thoracic, 80.35 % cervical. The evidence brief does not state how many distinct spines the 950 CTSpinoPelvic1K walks span, and that is not recorded here.

Those figures sit four to five points below the counting arms measured on the same protocol, which is the paper's central comparison: the learned step is the worse step.

What it costs to run

Table 3 gives the budget. The encoder takes 17–18 ms per frame, linking a whole walk takes 0.102–0.29 s, and decoding takes 0.2 ms. Against that, repositioning a mobile C-arm takes 36–49 s per move in the only published time-motion measurements [31, 32], and beam-on time per view is 0.5–4.7 s, about 11 % of positioning time [32]. A two-exposure walk needs one move beyond the first placement. The compute is not the constraint and no speed claim is made: the comparison a reviewer should apply is against 4.7 s to 72 s per image for the clinically translated registration system [22, 33], against which surgeons said they would wait 20 s to one minute [34].

Discussion

The result that matters is a division of labour. Counting across overlapping fields is exact and contains no learned component: 99.17–100 % of reached walks in nine arms of 16 to 801 studies each, at 1.15 to 2.96 exposures, with the annotation's boxes. Naming the same chains with a trained network was 94.7–96.3 %. The two numbers come from the same images, the same protocol and the same run, and the learned step is the weaker one. Converting an ordinal into a name requires an alphabet, and transitional anatomy changes the alphabet; a network's proper job here is not to recognise each vertebra from its morphology but to decide which alphabet the patient has. When a surgical plan is stated relative to the anchor—the third mobile segment above the sacrum rather than L4—counting alone suffices, and nothing needs to be learned at all.

What has to be true in the room. Three things. The anchor must be in the first field, which the radiographer controls. The vertebrae must be found in every frame, which a detector controls and which is where all of the residual risk now sits: substituting a detector's boxes for the annotation's dropped exactness from 100 % to 30.06 % on the same 79 spines, and the anchor node was correct on fewer than half the walks. And the patient must not move between exposures, which is not measured here at all.

What happens when it fails, and what it refuses. A wrong join or a wrong anchor shifts the count by one body and renames the column above it confidently. The mitigation designed for this is refusal—identify the anchor by class and decline when no anchor is found—and, as reported above, refusal fired zero times in every arm, so it is a design and not a result. The second mitigation is that the failure is auditable by eye: the chain is a strip of overlapping radiographs, and a surgeon can see whether the bodies shared between two frames are the same bodies. That is a property registration does not have, because a registration failure looks like a correctly placed overlay.

What it does not do. It forms no level name, so it cannot tell a surgeon that a body is L5 rather than L6; that question belongs to the naming model and is open. It does not reach targets the image does not cover, and 97 and 168 walks in two corpora ended for that reason. It was not measured on a moving patient, on parallax between exposures taken at different gantry angles, or in any operating room; every sweep in this paper was cut from a single long image. There is no reader study and no prospective arm.

Why this is deployable where navigation is not. The mechanism needs an anchor, two to three exposures on a C-arm the hospital already owns, and an embedded computer: 17–18 ms per frame is not a workstation-class load. It needs no preoperative CT, no optical tracker and no line of sight, and it asks for one model rather than two, because the count requires none. This is the setting in which wrong-level surgery is least protected—community hospitals and ambulatory surgery centres that do not buy navigation suites—but the capital cost of those suites was not measured in this study and no saving is claimed from avoiding one. Against the alternative that is priced, a preoperative gold fiducial costs 51.9 minutes and 183 mGy per patient [17], and a two-exposure sweep costs one C-arm move.

Where it belongs in the Universal Protocol. The protocol verifies a mark; this supplies the

spine's missing one. Verification enters the surgical field with the patient, the sweep places a digital mark on the target, and the time-out compares that mark's count against the surgeon's own. It is assistance and not replacement: two independent counts that must agree, with the disagreement displayed. A quality programme can audit it, because the evidence is a saved strip of images rather than a number a black box emitted.

Limitations. Every headline number uses the annotation's vertebra boxes, which measures the linker and not a deployable system; the detector-box arms are the honest preview and they are poor. The anchor-by-class arms that are meant to fix them have not returned. Sweeps are simulated from single long images and reproduce independent exposure, noise, rotation and scale between frames but not parallax and not patient motion, both of which act on linking. The correlation search is restricted to within 0.6 of a body of the commanded step, and in simulation the step taken is the step commanded, by construction; a radiographer's move is not, and the width of that discrepancy has not been measured. The real-radiograph arms rest on 16 and 24 spines. Refusal never fired. Two corpora—CTSpinoPelvic1K and VerSe—overlap with the training data of the naming model reported for comparison, although not of the count, which is fitted on nothing. TotalSegmentator carries no variant lumbar or thoracic class, so nothing here bears on transitional anatomy. The cost figures in the Introduction are extrapolations from verified rates onto an unverified surgical volume, and are stated as ranges for that reason.

Tables

Table 1: Counting arms with the annotation's vertebra boxes. No learned component takes part, so no spine is withheld. *Reached* excludes walks whose target never entered a field. *Wrong* gives walks with distinct spines in parentheses. The anchor diagnostic was 1.0000 in every row except the 522-spine sacrum arm (0.9935). Refusals were zero in every row. Transcribed from `docs/COUNTING_EVIDENCE.md`.

Anchor	Corpus	Spines	Walks	Reached	Exact	Exp.	Wrong
C2, 2 bodies	VerSe	76	834	834	99.76 %	1.99	2 (1)
C2, 3 bodies	VerSe	76	834	834	100 %	2.24	0 (0)
C2, 2 bodies	TotalSegmentator	204	2,656	2,654	99.70 %	2.05	8 (2)
C2, 3 bodies	TotalSegmentator	204	2,656	2,650	99.62 %	2.27	10 (2)
C2, 2 bodies	Real radiographs	24	110	110	100 %	1.15	0 (0)
Sacrum, 2 bodies	CTSpinoPelvic1K	801	6,352	6,255	99.66 %	1.98	21 (6)
Sacrum, 3 bodies	CTSpinoPelvic1K	801	6,352	6,229	99.76 %	2.42	15 (5)
Sacrum, 2 bodies	TotalSegmentator	522	7,396	7,228	99.17 %	2.96	60 (6)
Sacrum, 2 bodies	Real radiographs	16	95	95	100 %	1.61	0 (0)

Table 2: Where the residual risk is. The first three rows are the same 79 CTSpinoPelvic1K spines, held out from the detector, counted from the sacrum with three different box sources. *Anchor* is the share of walks whose starting node really was the sacrum. The film row's exactness and anchor diagnostic are not reconciled in this study. Transcribed from `docs/COUNTING_EVIDENCE.md`.

Boxes	Overlap	Spines	Reached	Exact	Anchor	Wrong
Annotation	2 bodies	79	618	100 %	100 %	0 (0)
Render detector	2 bodies	79	622	30.06 %	42.44 %	435 (79)
Render detector	3 bodies	79	621	28.66 %	39.77 %	443 (75)
Film detector	2 bodies	16	95	100 %	93.68 %	0 (0)

Table 3: The deployment budget. Rows without a citation were measured in this study's own runs. The compute is three orders of magnitude below one C-arm move, so the argument is not speed; it is that the count needs no model and the hardware is a C-arm the hospital already owns.

Quantity	Value	Source
Encoder, per frame (naming only)	17–18 ms	this study
Frame linking, per walk	0.102–0.29 s	this study
Chain decode, per walk	0.2 ms	this study
Exposures per walk	1.15–2.96	this study
C-arm reposition, per move	36–49 s	[31, 32]
Beam-on, per view	0.5–4.7 s	[32]
Registration labelling, per image	4.7–72 s	[22, 33]
Surgeon-stated acceptable wait	20 s–1 min	[34]
Thoracic localisation, fluoroscopy	$\approx$15 min	[19, 20]
Thoracic localisation, with fiducial	$\approx$3 min	[19, 20]
Fiducial placement itself	51.9 min, 183 mGy	[17]
Fiducial CT-suite intervention	14.3 min	[21]

References

[1] Michael J. Strong, Jacob Santarosa, Timothy P. Sullivan, Noojan Kazemi, Jacob R. Joseph, Osama N. Kashlan, Mark E. Oppenlander, Nicholas J. Szerlip, Paul Park, and Vincent Chang. Pre- and intraoperative thoracic spine localization techniques: a systematic review. *Journal of Neurosurgery: Spine*, 36(5):792–799, 2021. doi: 10.3171/2021.8.SPINE21480.

[2] J. E. Mayer, R. P. Dang, G. F. Duarte Prieto, S. K. Cho, S. A. Qureshi, and A. C. Hecht. Analysis of the techniques for thoracic- and lumbar-level localization during posterior spine surgery and the occurrence of wrong-level surgery: results from a national survey. *Spine J*, 14:741–748, 2014. doi: 10.1016/j.spinee.2013.06.068.

[3] Mary R. Kwaan, David M. Studdert, Michael J. Zinner, and Atul A. Gawande. Incidence, patterns, and prevention of wrong-site surgery. *Archives of Surgery*, 141(4):353–358, 2006. doi: 10.1001/archsurg.141.4.353.

[4] Philip F. Stahel, Allison L. Sabel, Michael S. Victoroff, Jeffrey Varnell, Alan Lembitz, Dennis J. Boyle, Ted J. Clarke, Wade R. Smith, and Philip S. Mehler. Wrong-site and wrong-patient procedures in the Universal Protocol era: analysis of a prospective database of physician self-reported occurrences. *Archives of Surgery*, 145(10):978–984, 2010. doi: 10.1001/archsurg.2010.185.

[5] Umile Giuseppe Longo, Mattia Loppini, Giacomo Romeo, Nicola Maffulli, and Vincenzo Denaro. Errors of level in spinal surgery: an evidence-based systematic review. *The Journal of Bone and Joint Surgery. British volume*, 94-B(11):1546–1550, 2012. doi: 10.1302/0301-620X.94B11.29553.

[6] Jon Devine, Norman Chutkan, Daniel C. Norvell, and Joseph R. Dettori. Avoiding wrong site surgery: a systematic review. *Spine*, 35(9 Suppl):S28–S36, 2010. doi: 10.1097/BRS.0b013e3181d833ac.

[7] Masayuki Furuya, Takahito Fujimori, Shota Takenaka, et al. Incidence and causes of wrong-level spine surgery: a multicentre registry of 34,544 degenerative spine operations. *The Bone & Joint Journal*, 108-B(5):673–680, 2026. doi: 10.1302/0301-620X.108B5.BJJ-2025-1129.R2.

[8] M. G. Mody, A. Nourbakhsh, D. L. Stahl, M. Gibbs, M. Alfawareh, and K. J. Garges. The prevalence of wrong level surgery among spine surgeons. *Spine*, 33(2):194–198, 2008. doi: 10.1097/BRS.0b013e31816043d1.

[9] Michael W. Groff, Joshua E. Heller, Eric A. Potts, Praveen V. Mummaneni, Christopher I. Shaffrey, and Justin S. Smith. A survey-based study of wrong-level lumbar spine surgery: the scope of the problem and current practices in place to help avoid these errors. *World Neurosurgery*, 79(3–4):585–592, 2013. doi: 10.1016/j.wneu.2012.03.017.

[10] Robert Goodkin and Lewis L. Laska. Wrong disc space level surgery: medicolegal implications. *Surgical Neurology*, 61(4):323–327, 2004. doi: 10.1016/j.surneu.2003.08.022.

[11] T. Bettis, Q. Tran, L. Glosser, F. Maegawa, and S. Patel. Malpractice payments for surgical never events reported to the National Practitioner Data Bank, 1991–2025. *Patient Safety in Surgery*, 2026. doi: 10.1186/s13037-026-00508-9.

[12] J. T. Machin, J. Hardman, W. Harrison, T. W. R. Briggs, and M. Hutton. Can spinal surgery in England be saved from litigation: a review of 978 clinical negligence claims against the NHS. *European Spine Journal*, 27(11):2693–2699, 2018. doi: 10.1007/s00586-018-5739-1.

[13] The Joint Commission. Sentinel event data: 2024 annual review. Oakbrook Terrace, IL: The Joint Commission, 2025. 127 wrong-surgery reports in 2024; voluntary reporting.

[14] NHS England. Provisional publication of never events reported as occurring between 1 April 2024 and 31 March 2025. NHS England, provisional StEIS data, england.nhs.uk, 2025.

[15] Winta T. Mehtsun, Andrew M. Ibrahim, Marie Diener-West, Peter J. Pronovost, and Martin A. Makary. Surgical never events in the United States. *Surgery*, 153(4):465–472, 2013. doi: 10.1016/j.surg.2012.10.005.

[16] Joy Tan, Jacqueline M. Ross, Daniel Wright, Marc Philip T. Pimentel, and Richard D. Urman. A contemporary analysis of closed claims related to wrong-site surgery. *The Joint Commission Journal on Quality and Patient Safety*, 49(5):265–273, 2023. PMID 36925434.

[17] Cole P. Siminski, Waleed Brinjikji, Naveen S. Murthy, Vance T. Lehman, Dong Kun Kim, Mohamad Bydon, Ahmad Nassr, Brett A. Freedman, Arjun S. Sebastian, and Felix E. Diehn.

Fluoroscopy- and CT-guided gold fiducial marker placement for intraoperative localization during spinal surgery: review of 179 cases at a single institution—technique and safety profile. *American Journal of Neuroradiology*, 44(5):618–622, 2023. doi: 10.3174/ajnr.A7855.

[18] Christopher P. Childers and Melinda Maggard-Gibbons. Understanding costs of care in the operating room. *JAMA Surgery*, 153(4):e176233, 2018. doi: 10.1001/jamasurg.2017.6233.

[19] Lorenzo Mongardi, Jacopo Visani, Giorgio Mantovani, et al. Preoperative CT-guided localization of thoracic spinal lesions for surgical planning. *see PMID 33358493*, 2021. PMID 33358493; venue and pages to be completed from the record.

[20] Cheerag D. Upadhyaya, Jau-Ching Wu, Cynthia T. Chin, G. Balamurali, and Praveen V. Mummaneni. Avoidance of wrong-level thoracic spine surgery: intraoperative localization with preoperative percutaneous fiducial screw placement. *Journal of Neurosurgery: Spine*, 16(3): 280–284, 2012. doi: 10.3171/2011.3.SPINE10445.

[21] Fee Keil et al. Preoperative CT-guided fiducial marker placement for intraoperative level localization in thoracic spine surgery. *see PMC11476588*, 2024. PMC11476588.

[22] Y. Otake, S. Schafer, J. W. Stayman, W. Zbijewski, G. Kleinszig, R. Graumann, A. J. Khanna, and J. H. Siewerdsen. Automatic localization of vertebral levels in x-ray fluoroscopy using 3d–2d registration: a tool to reduce wrong-site surgery. *Phys Med Biol*, 57:5485–5508, 2012. doi: 10.1088/0031-9155/57/17/5485.

[23] Y. Otake, A. S. Wang, A. Uneri, G. Kleinszig, S. Vogt, N. Aygun, S. F. Lo, J. P. Wolinsky, Z. L. Gokaslan, and J. H. Siewerdsen. 3d–2d registration in mobile radiographs: algorithm development and preliminary clinical evaluation. *Phys Med Biol*, 60:2075–2090, 2015. doi: 10.1088/0031-9155/60/5/2075.

[24] G. Schwing and OpenSpineConsortium. CTSpinoPelvic1K: 802 abdominopelvic CT records with per-level vertebrae, ribs, sacrum, hips and femora. Zenodo, concept DOI 10.5281/zenodo.22139642; Hugging Face gschwing/CTSpinoPelvic1K, 2026.

[25] A. Sekuboyina, M. E. Husseini, A. Bayat, M. Löffler, H. Liebl, H. Li, G. Tetteh, J. Kukačka,

C. Payer, D. Štern, et al. VerSe: a vertebrae labelling and segmentation benchmark for multi-detector CT images. *Med Image Anal*, 73:102166, 2021. doi: 10.1016/j.media.2021.102166.

[26] G. Schwing. Fixed-scale context crops with a target-marker channel for vertebral level identification. spinesurg-ct-nnunet, tools/make_context_crops.py; docs in repository, 2026.

[27] A. Boduch, M. Hennrikus, T. Adebayo, J. Brian, and W. Hennrikus. Accuracy of C-arm measurements in assessment of paediatric femoral fracture shortening. *Journal of Children's Orthopaedics*, 11(6):449–453, 2017. doi: 10.1302/1863-2548.11.170116.

[28] Yoshito Otake, Sebastian Schafer, J. Webster Stayman, Wojciech Zbijewski, Gerhard Kleinszig, Rainer Graumann, A. Jay Khanna, and Jeffrey H. Siewerdsen. Automatic localization of vertebral levels in x-ray fluoroscopy using 3D-2D registration: a tool to reduce wrong-site surgery. *Physics in Medicine and Biology*, 57(17):5485–5508, 2012. doi: 10.1088/0031-9155/57/17/5485.

[29] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *CVPR*, pages 770–778, 2016.

[30] G. Schwing. Mixed-length training of a run-level recurrent head over vertebra embeddings. spinesurg-ct-nnunet, tools/head_mixed_length.py; results in model_night/heads_mixed, 2026.

[31] Felix Matthews, Dominik J. Hoigne, Martin Weiser, Guido A. Wanner, Pietro Regazzoni, Norbert Suhm, and Peter Messmer. Navigating the fluoroscope's C-arm back into position: an accurate and practicable solution to cut radiation and optimize intraoperative workflow. *Journal of Orthopaedic Trauma*, 21(10):687–692, 2007. doi: 10.1097/BOT.0b013e318158fd42.

[32] Tharindu De Silva, Joshua Punnoose, Ali Uneri, Mahadevappa Mahesh, Joseph Goerres, Matthew Jacobson, Michael D. Ketcha, Amir Manbachi, Sebastian Vogt, Gerhard Kleinszig, A. Jay Khanna, Jean-Paul Wolinsky, Greg Osgood, and Jeffrey H. Siewerdsen. Virtual fluoroscopy for intraoperative C-arm positioning and radiation dose reduction. *Journal of Medical Imaging*, 5(1):015005, 2018. doi: 10.1117/1.JMI.5.1.015005.

[33] A. Manbachi, T. De Silva, A. Uneri, M. Jacobson, J. Goerres, M. Ketcha, R. Han, N. Aygun, D. Thompson, X. Ye, S. Vogt, G. Kleinszig, C. Molina, D. M. Sciubba, J.-P. Wolinsky, and J. H. Siewerdsen. Clinical translation of the LevelCheck decision support algorithm for target localization in spine surgery. *Ann Biomed Eng*, 46(10):1548–1557, 2018. doi: 10.1007/s10439-018-2099-2.

[34] Sheng-fu L. Lo, Tao Yi, Yoshito Otake, Ali Uneri, Dominic Reh, Ziya L. Gokaslan, Jeffrey H. Siewerdsen, and Jean-Paul Wolinsky. Utility of the LevelCheck algorithm for decision support in vertebral localization. *Spine*, 41(21):E1279–E1283, 2016. doi: 10.1097/BRS.0000000000001589. PMID 27035579. The 42.2% and 30.6% are questionnaire ratings, the mean of three surgeons rating independently, range 31.7–53.3%; the companion 100% is geometric label placement conditioned on a correctly annotated CT.